

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

The volume V of an enclosure contains a mixture of three gases, 16 g of oxygen, 28 g of nitrogen and 44 g of carbon dioxide at absolute temperature T . Consider R as universal gas constant. The pressure of the mixture of gases is :

- (1) $\frac{88RT}{V}$
- (2) $\frac{3RT}{V}$
- (3) $\frac{5}{2} \frac{RT}{V}$
- (4) $\frac{4RT}{V}$

Q2: 16 March (Shift 2) - Single Correct

Calculate the value of mean free path (λ) for oxygen molecules at temperature 27°C and pressure $1.01 \times 10^5 \text{ Pa}$. Assume the molecular diameter 0.3nm and the gas is ideal.

$$(k = 1.38 \times 10^{-23} \text{ JK}^{-1})$$

- (1) 58nm
- (2) 32nm
- (3) 86nm
- (4) 102nm

Q3: 17 March (Shift 1) - Single Correct

A polyatomic ideal gas has 24 vibrational modes. What is the value of γ ?

- (1) 1.03
- (2) 1.3
- (3) 1.37
- (4) 10.3

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Q4: 17 March (Shift 1) - Single Correct

Two ideal polyatomic gases at temperatures T_1 and T_2 are mixed so that there is no loss of energy. If F_1 and F_2 , m_1 and m_2 , n_1 and n_2 be the degrees of freedom, masses, number of molecules of the first and second gas respectively, the temperature of mixture of

these two gases is :

- (1) $\frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$
- (2) $\frac{n_1 F_1 T_1 + n_2 F_2 T_2}{n_1 F_1 + n_2 F_2}$
- (3) $\frac{n_1 F_1 T_1 + n_2 F_2 T_2}{F_1 + F_2}$
- (4) $\frac{n_1 F_1 T_1 + n_2 F_2 T_2}{n_1 + n_2}$

Q5: 17 March (Shift 2) - Single Correct

If one mole of the polyatomic gas is having two vibrational modes and β is the ratio of molar specific heats for polyatomic gas $\left(\beta = \frac{C_p}{C_v}\right)$ then

the value of β is :

- (1) 1.02
- (2) 1.2
- (3) 1.25
- (4) 1.35

Q6: 18 March (Shift 1) - Single Correct

What will be the average value of energy along one degree of freedom for an ideal gas in thermal equilibrium at a temperature T (k_B is Boltzmann constant)

- (1) $\frac{1}{2} k_B T$
- (2) $\frac{2}{3} k_B T$
- (3) $\frac{3}{2} k_B T$
- (4) $k_B T$

Q7: 18 March (Shift 2) - Single Correct

Consider a sample of oxygen behaving like an ideal gas. At 300 K, the ratio of root mean square (rms) velocity to the average velocity of gas molecule would be :

(Molecular weight of oxygen is 32 g/mol; $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$)

(1) $\sqrt{\frac{3}{8}}$

(2) $\sqrt{\frac{8}{3}}$

(3) $\sqrt{\frac{3\pi}{8}}$

(4) $\sqrt{\frac{8\pi}{3}}$

Answer Key**Q1 (3)****Q2 (4)****Q3 (1)****Q4 (2)****Q5 (2)****Q6 (1)****Q7 (3)**

Q1 (3)

$$PV = (n_1 + n_2 + n_3) RT$$

$$P \times V = \left[\frac{16}{32} + \frac{28}{28} + \frac{44}{44} \right] RT$$

$$PV = \left[\frac{1}{2} + 1 + 1 \right] RT$$

Q2 (4)

$$\lambda = \frac{RT}{\sqrt{2}\pi d^2 N_A P}$$

$$\lambda = 102 \text{ nm}$$

Q3 (1)

Since each vibrational mode has 2 degrees of freedom hence total vibrational degrees of freedom = 48

$$f = 3 + 3 + 48 = 54$$

$$\gamma = 1 + \frac{2}{f} = \frac{28}{27} = 1.03$$

Q4 (2)

Let the final temperature of the mixture be T. Since, there is no loss in energy,

$$\Delta U = 0$$

$$\Rightarrow \frac{F_1}{2} n_1 R \Delta T + \frac{F_2}{2} n_2 R \Delta T = 0$$

$$\Rightarrow \frac{F_1}{2} n_1 R (T_1 - T) + \frac{F_2}{2} n_2 R (T_2 - T) = 0$$

$$\Rightarrow T = \frac{F_1 n_1 R T_1 + F_2 n_2 R T_2}{F_1 n_1 R + F_2 n_2 R} \Rightarrow \frac{F_1 n_1 T_1 + F_2 n_2 T_2}{F_1 n_1 + F_2 n_2}$$

Q5 (2)

$$f = 4 + 3 + 3 = 10$$

assuming non linear

$$\beta = \frac{C_p}{C_v} = 1 + \frac{2}{f} = \frac{12}{10} = 1.2$$

Q6 (1)

Energy associated with each degree of freedom per molecule = $\frac{1}{2}k_B T$.

Q7 (3)

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$
$$V_{avg} = \sqrt{\frac{8}{\pi} \frac{RT}{M}}$$
$$\frac{v_{rms}}{V_{avg}} = \sqrt{\frac{3\pi}{8}}$$